

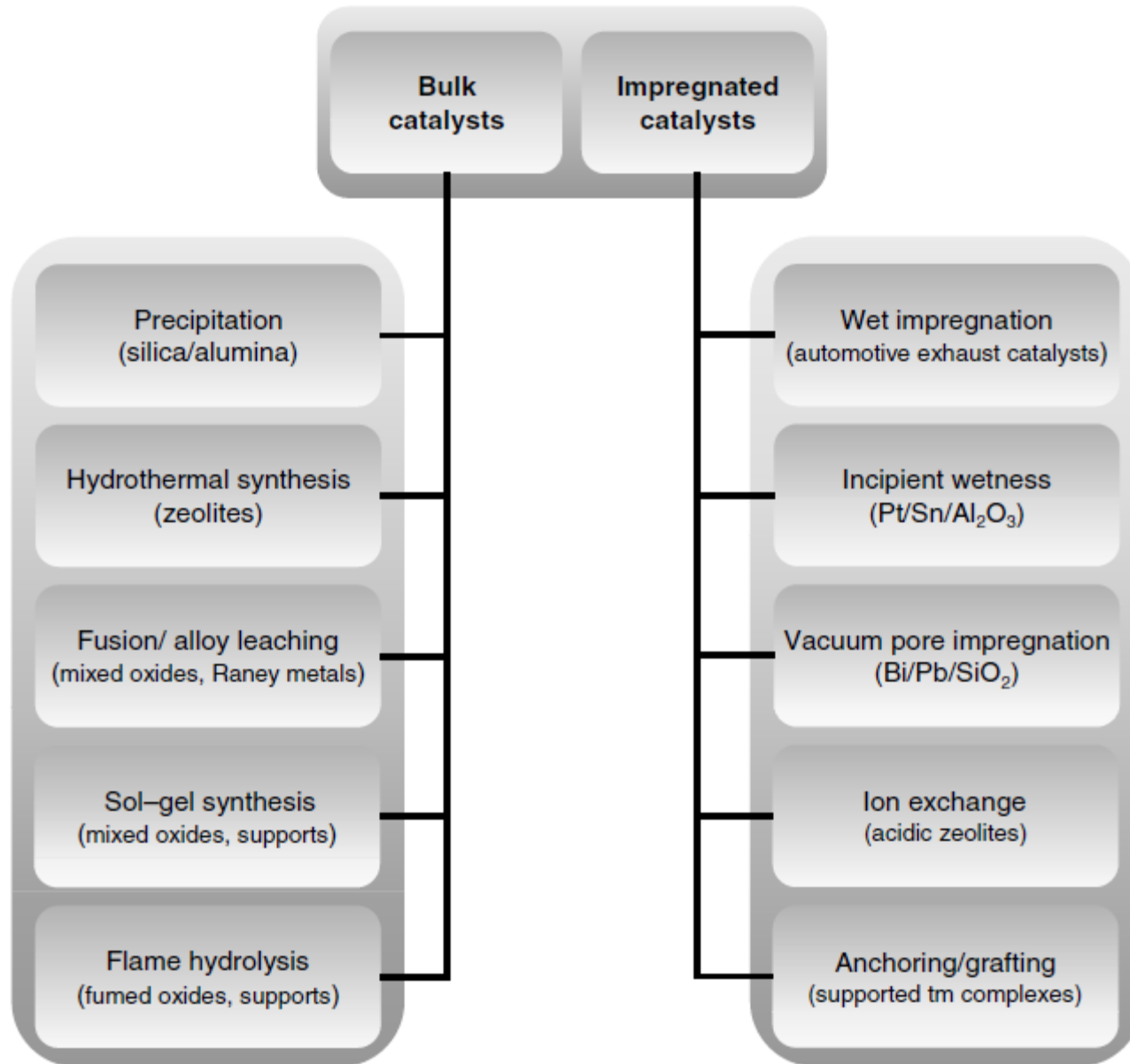


CHEM F337: Green Chemistry & Catalysis

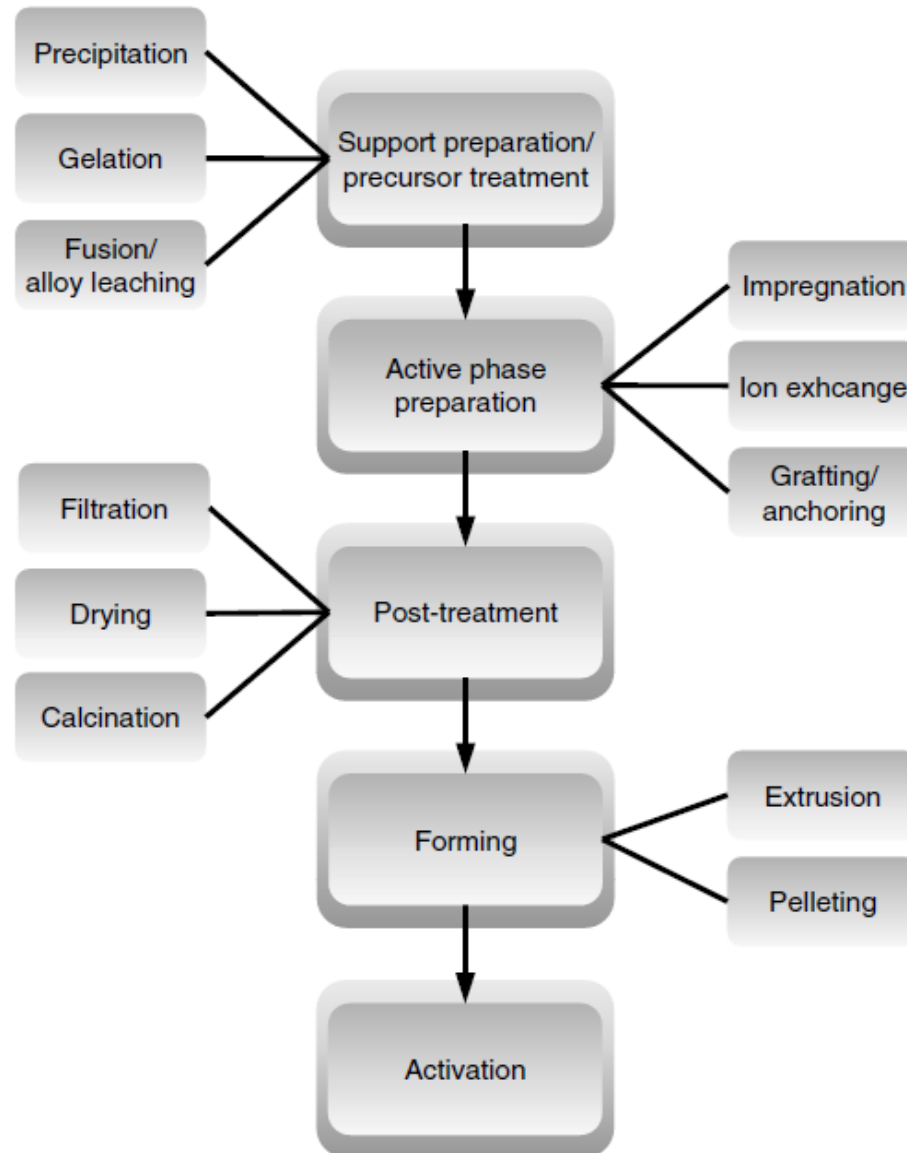
Lecture 24

Heterogeneous Catalysis: Synthesis

Solid Catalyst Preparation



Solid Catalyst Preparation - Large Scale Unit Operations



This flowchart is only for industrial scale

Solid Catalyst Preparation



Precipitation / co-precipitation method

Typically, the precursor solutions (usually aqueous salt solutions) are mixed, and the desired salt is precipitated as a gel by adding an acid/base or another reagent. The resulting gel is aged, filtered, washed, dried, and finally calcined

Precipitation has three stages: **supersaturation, nucleation, and growth**

Advantages

- Precipitation can give high-purity materials
- Co-precipitation can give stoichiometric mixtures with well-defined mixed crystallites.

Disadvantages

- Precipitation requires a solvent and a precipitating agent.
- Added costs for catalyst separation and waste disposal.

Solid Catalyst Preparation



High Temperature Fusion and Alloy Leaching Method

Metallic alloys and some mixed-oxide catalysts can be prepared by co-melting and fusion of the metal (oxide) precursors at high temperatures. Alloys and mixed-oxide catalysts are usually referred to by combining the formulas of their precursors with a dash, e.g. V_2O_5 - MoO_3 , or simply as V-Mo-O.

Advantages

- The melting enables intimate mixing of the precursor atoms and clusters, creating highly pure, well dispersed solids.
- Range of different phases can be obtained by controlling the cooling of the melt.

Disadvantages

- The disadvantage of the melting/fusion option is that it is an energy-intensive process which requires special equipment

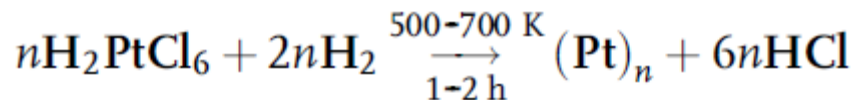
Solid Catalyst Preparation



Impregnation of porous/ non-porous supports

In wet impregnation, the (porous) support is immersed in a solution of the catalyst precursor. The precursor is either adsorbed spontaneously on the support, or precipitated there by changing the pH or by inducing another chemical reaction. The resulting catalyst is then filtered, dried, and calcined.

Pt/Al₂O₃ is prepared by mixing porous alumina with a chloroplatinic acid solution. The solvent is evaporated, leaving the solute dispersed on the surface. The catalyst is dried and the supported Pt^{IV} ions are reduced to Pt⁰ with hydrogen gas. (Pt)_n indicates a Pt cluster, (100-1000 atoms depending on the degree of aggregation during the reduction step)



Advantages

- High dispersion of the active metal can be obtained

Disadvantages

- Large volumes of aqueous waste involved

Solid Catalyst Preparation



Impregnation of porous/ non-porous supports

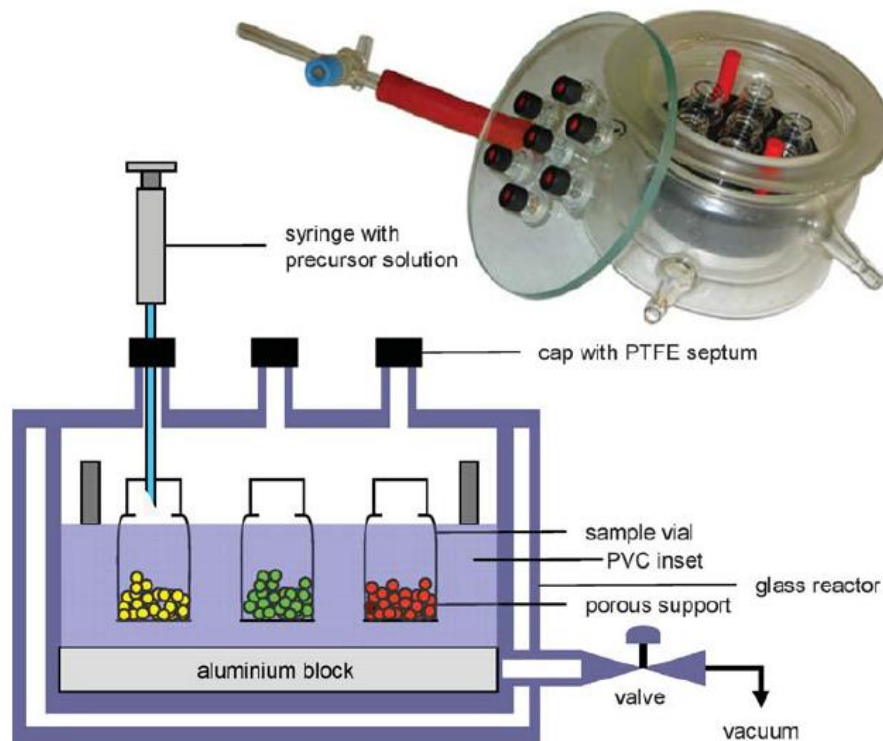
Bypasses to disadvantages

1. Incipient Wetness Technique:

solution of the active precursor is added to the dry support powder until the mixture becomes “slightly tacky”, indicating that the pores of the support are filled with the liquid.

1. Vacuum Pore Impregnation Technique:

- The support is first dried and then placed under vacuum, emptying the pores of air. A volume of precursor solution equivalent to the pore volume is then mixed with the support under vacuum, and this solution is absorbed in the pores of the seemingly-dry powder.
- High loadings of active material can be achieved by drying the catalyst and repeating the impregnation process.



Solid Catalyst Preparation



Hydrothermal method

Hydrothermal treatments involve synthesis using heat in presence of water – the heating of precipitates, gels, or flocculates in the presence of water.

These treatments are typically carried out in an autoclave at 100–300 C, and result in textural and/or structural changes, including **crystal and/or particle growth, changes in crystal structure, and transformation of amorphous solids into crystalline ones.**

The most important large-scale application of hydrothermal synthesis is the synthesis of **porous materials** such as **microporous, mesoporous and macroporous materials**. Others include **nanomaterials, electrocatalysts etc.**

What are microporous, mesoporous and macroporous materials?

Depending on the sizes of pores the porous materials can be called so

- Pores $< 20\text{\AA}$, $\Rightarrow$ Microporous materials, e.g. zeolites
- $20\text{\AA} < \text{Pores} < 200\text{\AA}$, $\Rightarrow$ Mesoporous materials, e.g. MCM-41, ALPOs, SAPOs
- Pores $> 200\text{\AA}$, $\Rightarrow$ Macroporous Materials, e.g. Clays

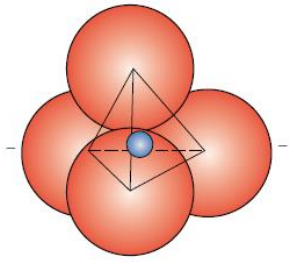
Majority of porous materials can act as sieves: called as Molecular Sieves.

All zeolites are molecular sieves while all molecular sieves are not zeolites

Solid Catalysts – Porous Matrices

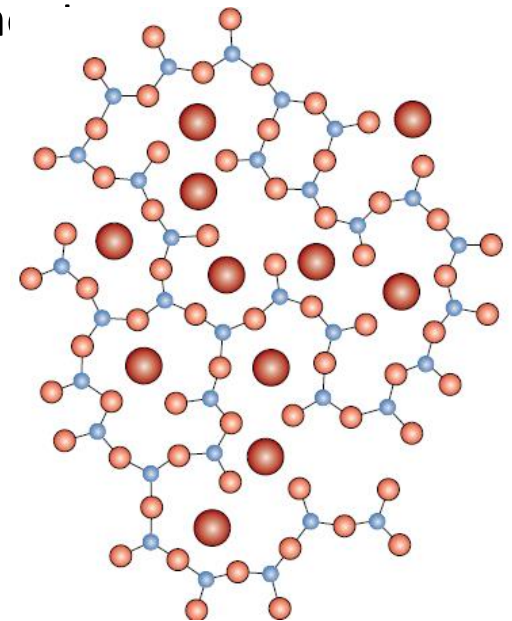


- Most important are silicates which are parts of all minerals obtained in Earth's crust, simplest being SiO_4^{4-} (orthosilicate, not that abundant as pure) - **basic unit of silicates (T units)**. Different arrangements of this unit will lead to different structures by one, two or three dimensional arrangement.



Silica

Chemically, the most simple silicate material is silicon dioxide, or silica (SiO_2). Structurally, a 3D network that is generated by corner sharing O atoms by adjacent tetrahedra. material is electrically neutral, stable, the ratio of Si to O atoms is 1:2



Simple silicates

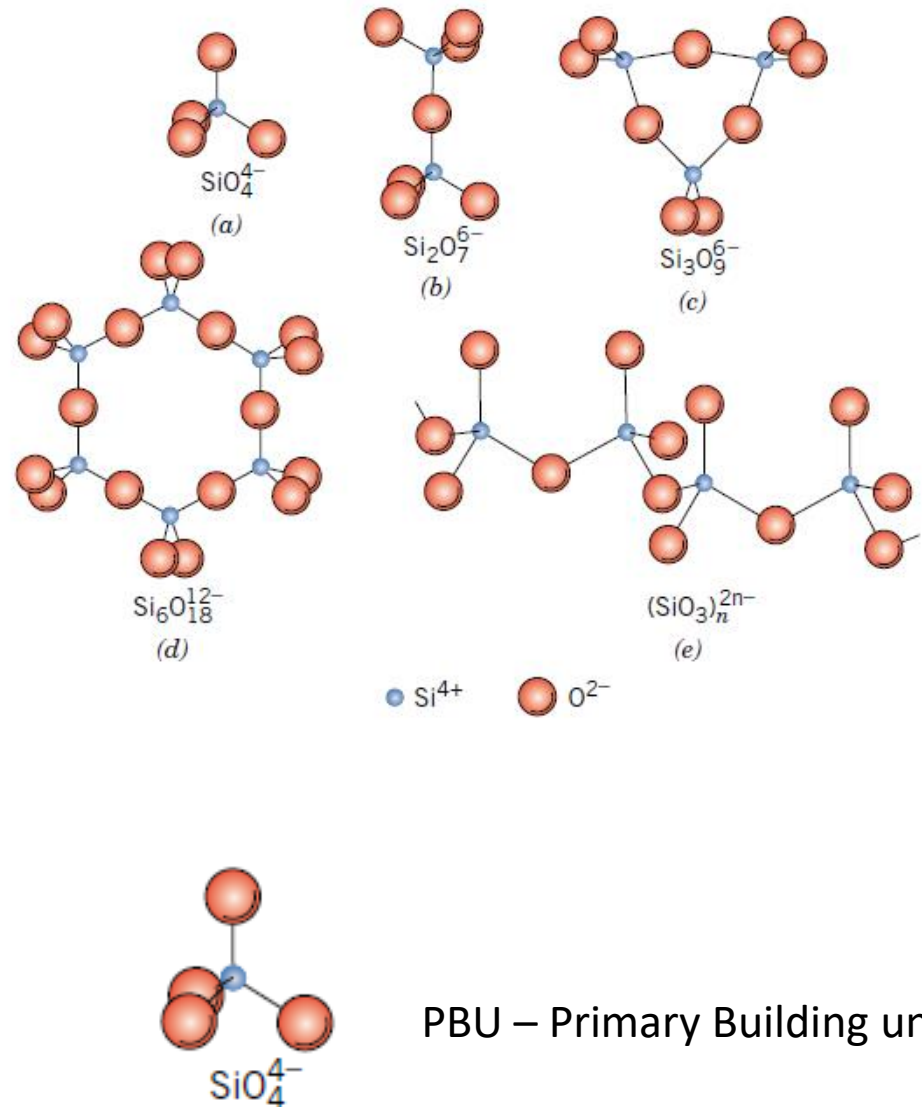
Of other silicates, the most structurally simple ones involve isolated tetrahedra (*E.g. forsterite (Mg_2SiO_4) has the equivalent of two Mg^{2+} ions associated with each tetrahedron in such a way that every Mg^{2+} ion has six oxygen nearest neighbors.*

Solid Catalysts – Porous Matrices



Other Silicates

- One, two, or three of the corner oxygen atoms of the SiO_4^{4-} tetrahedra are shared by other tetrahedra to form some rather complex structures.
- Some of these have formulas SiO_4^{4-} , $\text{Si}_2\text{O}_6^{6-}$, $\text{Si}_3\text{O}_9^{6-}$ and so on;
- Single-chain structures are also possible,
- *Positively* charged cations such as Ca^{2+} , Mg^{2+} , and Al^{3+} serve two roles: First, they compensate the negative charges from the SiO_4^{4-} units so that charge neutrality is achieved; second, these cations ionically bond the SiO_4^{4-} tetrahedra together.



Zeolite structures:

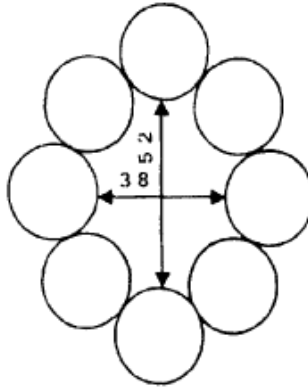
Zeolites are microporous aluminosilicates where tetrahedral Al^{3+} units are present along with Si^{4+} centers in the T-units and hence they require cations for maintaining electroneutrality

Their porous structures have TWO aspects –

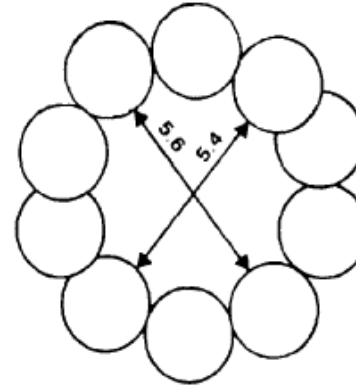
Hollows (cages) bound by **Windows (pore periphery)**

Cages & windows structure actually are responsible for their properties

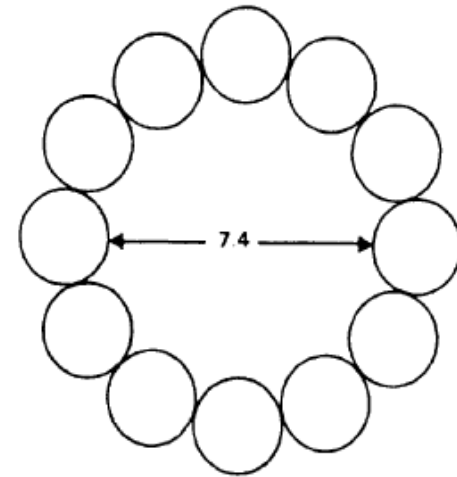
Erionite 8 Ring



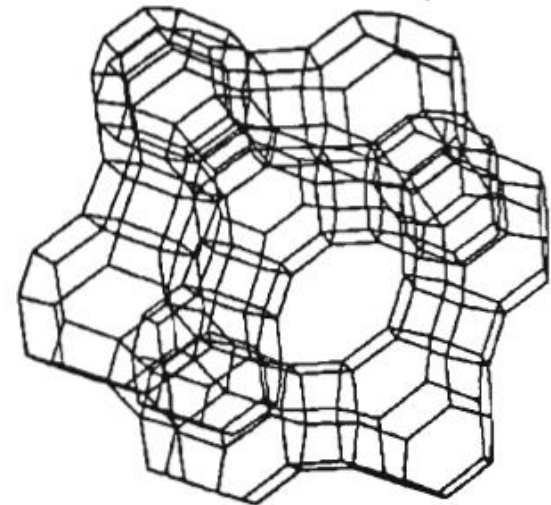
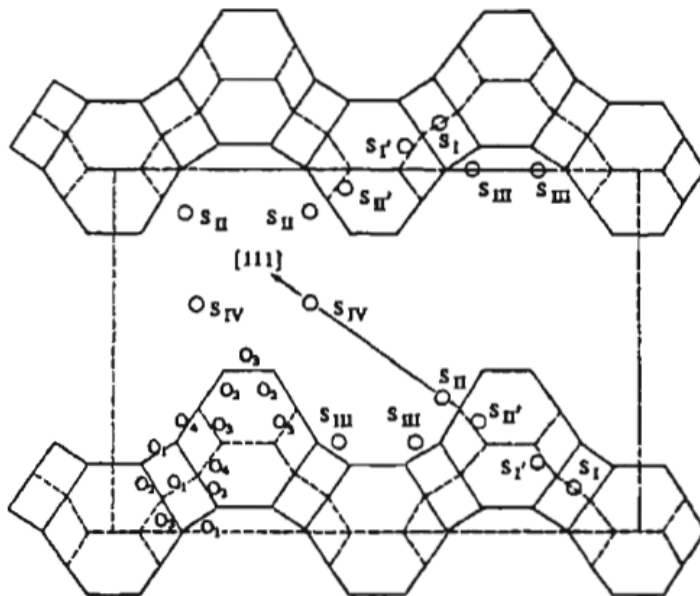
ZSM-5 10 Ring (Straight Channel)



Y Zeolite 12 Ring

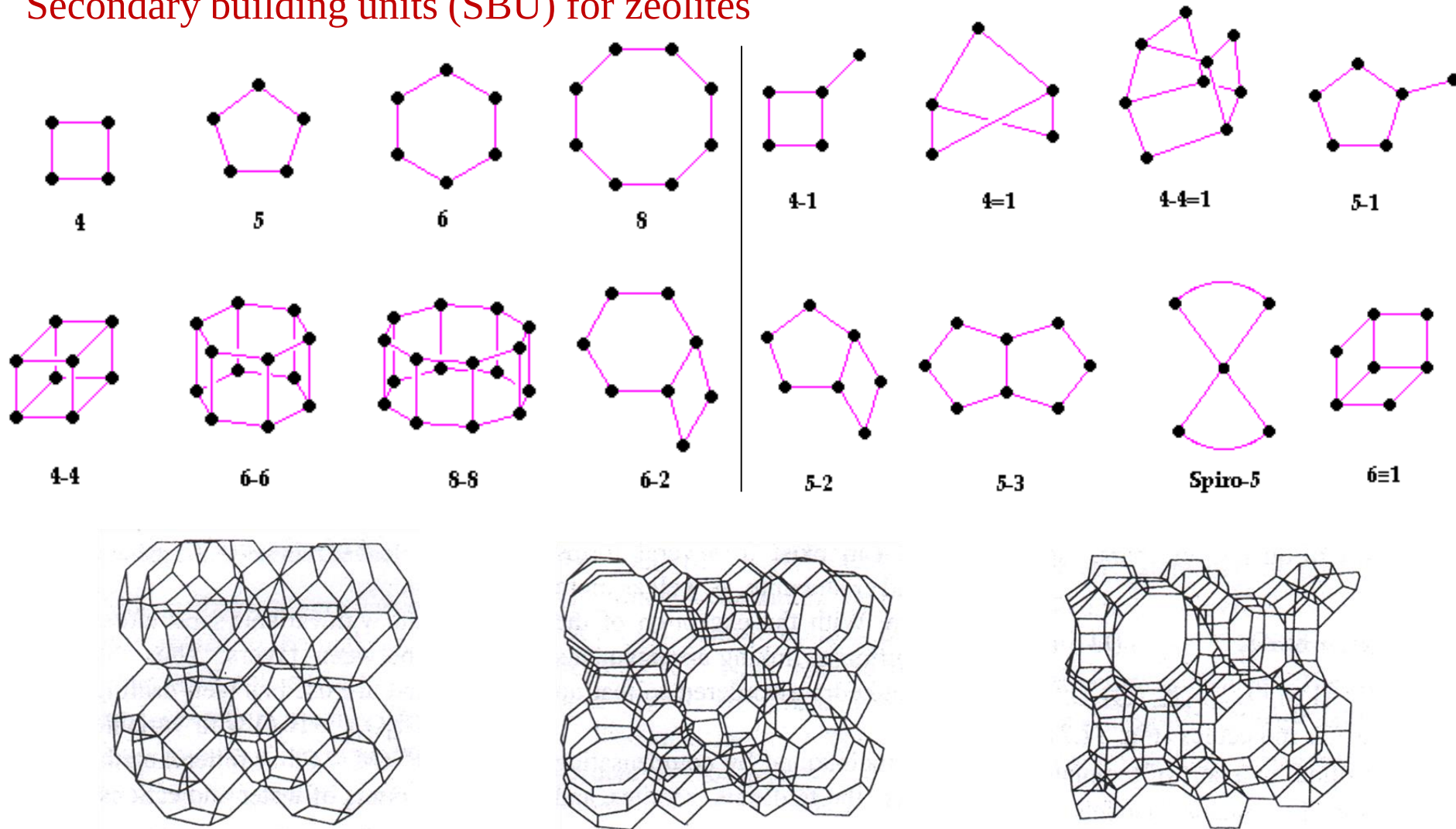


Sinusoidal Channel



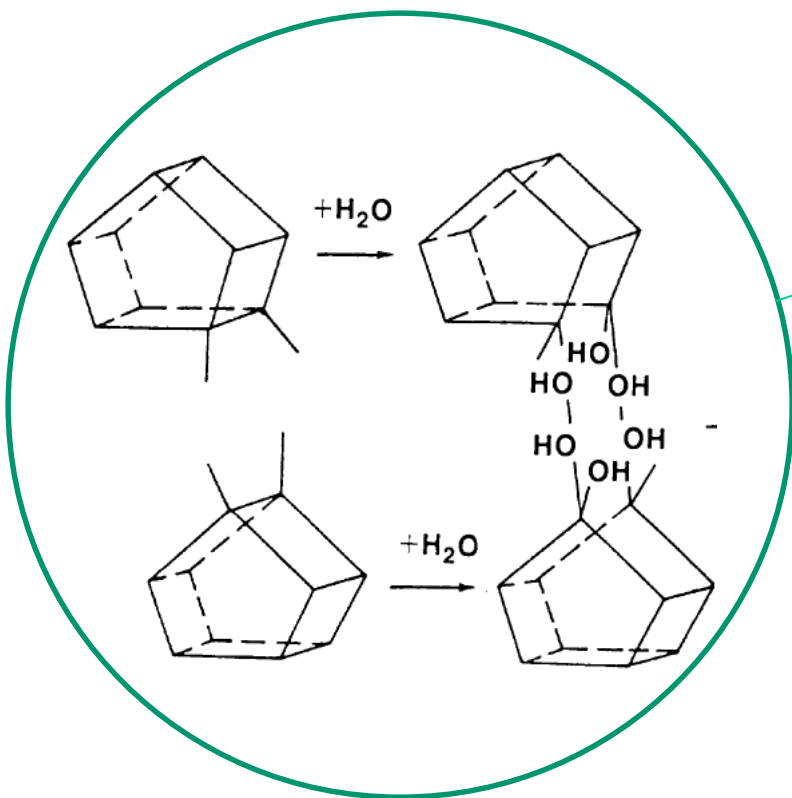
Silicates: SBUs

Secondary building units (SBU) for zeolites



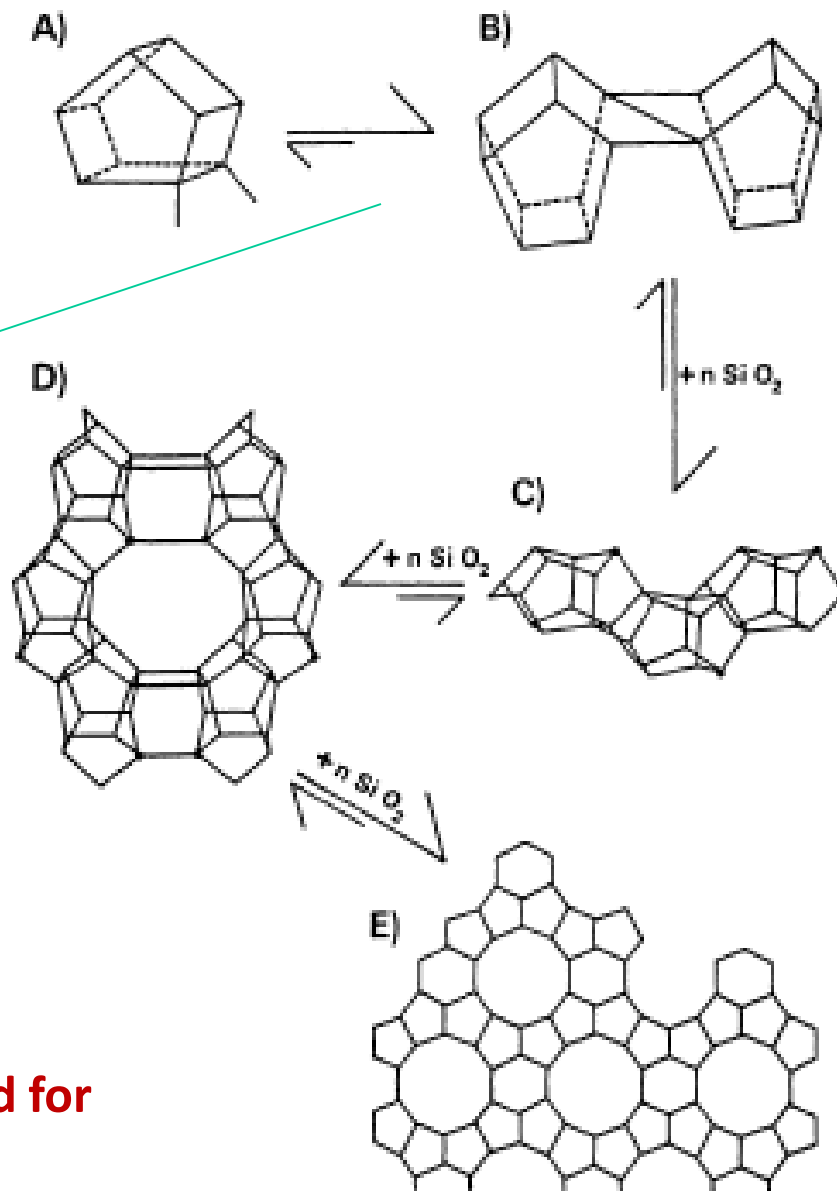
Complex structures developed during hydrothermal synthesis

Hydrothermal Synthesis - Zeolites

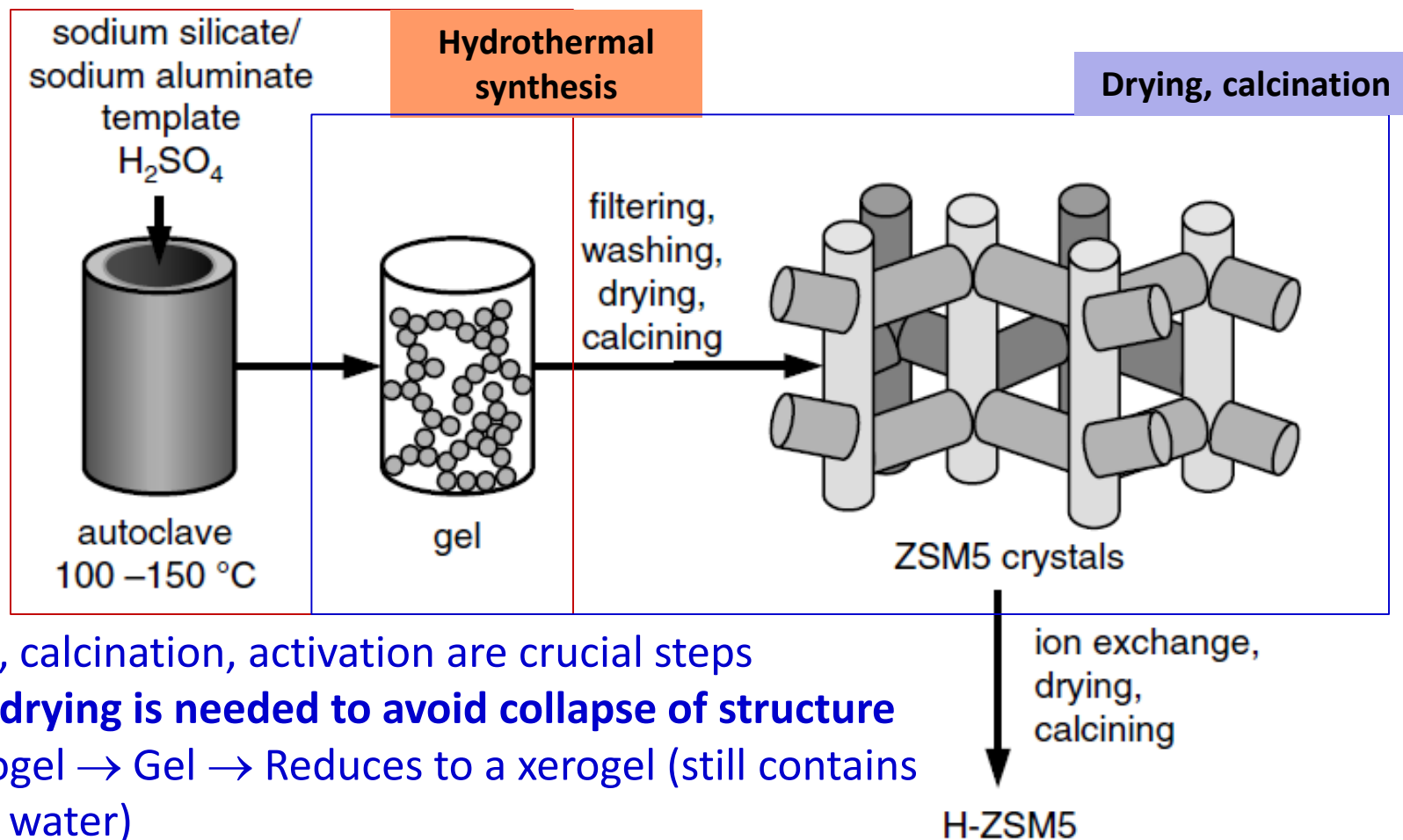


Stepwise Self Assembly
of ZSM-5 framework

Similar pathways are also followed for
AlPOs, SAPOs etc.



Hydrothermal Synthesis - Zeolites



Drying, calcination, activation are crucial steps

- **Slow drying is needed to avoid collapse of structure**
- Hydrogel $\rightarrow$ Gel $\rightarrow$ Reduces to a xerogel (still contains $\sim 30\%$ water)
- **Calcination (controlled)** removes all water from the material – important for generating **surface hydrophobicity**

The Si/Al ratio in the recipe usually determines surface acidity – **Bronsted & Lewis**
Exchangeable framework cations – **exchange capacity**